VATSAL MAURYA

vatsal.maurya@alumni.iiserpune.ac.in
Website | LinkedIn

Systems modelling for climate adaptation | Climate risk | Environmental data science

Research interests: climate adaptation and resilience; complex systems; climate-risk modelling; spatio-temporal dynamics; agent-based and network models.

CURRENT ENGAGEMENT

Climate Data Consultant (Part-time) Sep 2026–Jan 2027

Khushi Baby | Climate-health vulnerability and environmental data methodology

- Standardizing the methodology for environmental variables used in the Climate Health Vulnerability Index (CHVI).
- Assessing appropriate spatio-temporal aggregation scales for input data and documenting scale-dependent trade-offs for vulnerability analysis.

EDUCATION

Indian Institute of Science Education and Research Pune (IISER Pune) 2021–2026

Integrated BS-MS in Science | Applied Mathematics and Statistics | CGPA: 8.8/10

- BS-MS thesis: From Grid Cells to Policy in Heat-Stress Analysis; supervised by Joy M. Monteiro (IISER Pune) and Deepak Malghan (IIM Bangalore).
- Institute Best MS Thesis Award (Data Science).

WORKING PAPERS & ONGOING RESEARCH

Mind the (Jensen) Gap: Climate and Health using Gridded Data Mar 2026

Vatsal Maurya, Joy M. Monteiro, Anish Sugathan, Deepak Malghan | IIMB Working Paper No. 732/2026

- Develops a multiscale framework for nonlinear environmental metrics and threshold exceedances, showing how spatio-temporal aggregation can produce systematic, spatially clustered and threshold-sensitive bias.

Heat-stress metrics and the timing of peak season in India Ongoing

Manuscript in preparation

- Compare the annual contiguous 90-day windows that maximize air temperature, WBGT, UTCI and Extended Heat Index using hourly ERA5 data for 1980–2024; preliminary results show substantial timing disagreement despite strong hour-to-hour metric correlation.

Changing Temporal Organisation of Heat Events Across India Ongoing

Manuscript in preparation

- Using hourly UTCI over India (1980–2024), quantify first and last annual threshold exceedances, hot-season duration, heat-event frequency and length, nighttime exposure, and breaks between events to test whether climate change is reorganising event timing, persistence, and recovery.
- Current results show strongly spatially heterogeneous shifts, with evidence of earlier starts, later ends, longer events, or shorter recovery periods across large parts of India; North-East India, the eastern coast, and the Indo-Gangetic Plain recur across multiple indicators and thresholds.

Who Is an Outdoor Worker? Ongoing

Third author | Climate-data contribution

- Supplied climate data used to estimate harmful heat-stress exposure in an eight-country labour-force study; the study finds that outdoor-worker proxies can disagree by up to 44 percentage points and materially alter downstream health and economic risk estimates.

IMD heat-wave criteria through uncompensable heat strain Ongoing

Ongoing manuscript project

- Contributed the initial partitional-calorimetry implementation and exploratory analysis using HadISD station records for India, Pakistan and Bangladesh.
- Ongoing collaboration evaluates how temperature-based IMD heat-wave declarations align with physiological heat risk by comparing uncompensable heat-strain hours on declared and non-declared days; manuscript in preparation.

RESEARCH EXPERIENCE

BS-MS Thesis Research — From Grid Cells to Policy in Heat-Stress Analysis 2025–2026

IISER Pune & Indian Institute of Management Bangalore | Supervisors: Joy M. Monteiro and Deepak Malghan

- Built and analysed global hourly ERA5/ERA5-Land heat-stress data at native resolution, using parallel and chunked workflows to compute UTCI, WBGT and meteorological intermediates; the ERA5 archive used in the thesis spans 1980–2024 and ~11–12 TB.
- Developed a multiscale framework for translating grid-level climate hazards to administrative decision scales, quantifying Jensen gaps introduced by nonlinear aggregation and thresholding.
- Modelled long-run changes in the temporal organisation of heat risk across India, including shifts in hot-season onset, duration, event persistence and recovery breaks over 45 years.
- Fit spatially explicit gridpoint logistic models of changing heat-event regimes, estimating evolving contributions of temperature, humidity, radiation, wind and daytime conditions.

ADDITIONAL RESEARCH EXPERIENCE

Graph-theoretic analysis of road networks to identify critical road segments Jun 2024–Jan 2025

Online project, IIASA Vienna | Advisors: Jose Moran and Celian Colon

- Modelled road infrastructure as a network system and applied graph-theoretic and simulation-based methods in NetworkX to identify critical and vulnerable segments.

Human body energy model for physiologically dangerous heat events Aug 2023–Dec 2024

IISER Pune | Advisor: Joy M. Monteiro

- Implemented a partitional-calorimetry model in Python and evaluated it on station-level meteorological data; this work provided the computational foundation for the ongoing IMD heat-wave criteria project.

Agent-based models of economic mobility and inequality May–Jul 2022

ETC Lab, Jawaharlal Nehru University | Advisor: Anirban Chakraborti

- Implemented Schelling and kinetic wealth-exchange models, including hybrid variants, to study how local interaction rules generate system-level patterns of inequality and mobility.

TECHNICAL SKILLS

Programming: Python (primary), C++, Java
Scientific / statistical computing: statsmodels, PySAL/esda, pymannkendall, NumPy/SymPy, NetworkX
Geospatial / climate data: ERA5, ERA5-Land, HadISD, QGIS, NetCDF, Zarr; parallel and chunked large-scale data workflows
Modelling: spatio-temporal statistical modelling, logistic regression, network analysis, agent-based modelling, partitional calorimetry
Languages: English (professional), Hindi (native), Kannada (elementary)

SELECTED AWARDS & SCHOLARSHIPS

- KVPY-SX Academic Fellowship, 2021–2026.
- Institute Best MS Thesis Award (Data Science), IISER Pune.

OTHER LEADERSHIP

Sponsorship Coordinator — Karavaan 2022 Aug–Nov 2022

IISER Pune

- Led sponsorship outreach for the institute's annual cultural festival and coordinated a team across departments.